

SNTL-NTU DCASE25 SUBMISSION: ACOUSTIC SCENE CLASSIFICATION USING CNN-GRU MODEL WITHOUT KNOWLEDGE DISTILLATION

Technical Report

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ABSTRACT

In this technical report, we present the SNTL-NTU team's Task 1 submission for the Low-Complexity Acoustic Scene Classification of the Detection and Classification of Acoustic Scenes and Events (DCASE) 2025 challenge [1]. This submission departs from the typical application of knowledge distillation from a teacher to a student model, aiming to achieve high performance with limited complexity. The proposed model is based on a CNN-GRU model and is trained solely using the TAU Urban Acoustic Scene 2022 Mobile development dataset [2], without utilizing any external datasets, except for MicIRP [3], which is used for device impulse response (DIR) augmentation. Two models have been submitted to this challenge with memory usage not more than 117 KB and requiring 10.9M multiply-and-accumulate (MAC) operations. Using the development dataset, the proposed model achieved an accuracy of 60.25%.

Index Terms— Acoustic scene analysis, CNN-GRU

1. INTRODUCTION

In Task 1 of the DCASE Challenge 2025, acoustic scene classification (ASC) is employed to classify 10 acoustic scenes from 12 cities based on 1-second audio samples. To align ASC with the performance of typical edge devices, Task 1 [1] of the DCASE Challenge 2023 has imposed the following system complexity constraints:

- Maximum memory allowance: 128 KB
 - Maximum number of MACs per inference: 30 MMAC
- Numerous CNN models distilled from large pretrained teachers have dominated the submissions in DCASE Task 1 since these networks achieved the highest accuracies with the TAU Urban Acoustic Scene 2022 Mobile dataset [2] since 2022. Along with the knowledge distillation (KD), augmentation techniques were extensively applied to enhance the model's generalizability to unseen devices and the variability of the audio samples obtained for the same scenes from different cities. To further reduce the model size, post-training quantization and pruning were applied to the weights and parameters of models.

The proposed model is a shallow CNN network combining depthwise (DW) convolution, channel shuffle (CS), squeeze-and-excite (SE), and gated recurrent unit (GRU) to effectively perform ASC. The MAC of the proposed model is 10.9M, and 16-bit precision is used to quantize the proposed model since the number of parameters is less than 64K.

The remaining sections of this report are organized as follows. In Section 2, the input features, augmentation techniques used, and proposed model are discussed. Section 3 presents the results of our submissions based on the various splits of the development dataset. This report is then concluded in Section 4.

2. PROPOSED SYSTEM

2.1. Preprocessing

The TAU urban acoustic scene 2022 mobile dataset contains recordings of 10 acoustic scenes in 12 European cities. These recordings are captured using four devices and synthetic data for 11 devices was generated using the recordings. Each 1 sec audio sample is captured with a sampling frequency of 44.1 kHz sampling rate and encoded at 24-bit resolution.

For the input feature, a 256 bin mel-spectrogram is calculated using the short time Fourier transform (STFT) with a window length of 0.18 sec and 17% overlap. This configuration produces 33 time frames and an input feature shape of $[256 \times 33]$ for each audio sample.

The input features and the down-sampling of the audio samples are computed and performed using Librosa [3]. The sampling rate 44.1kHz is selected based on the observation on the averaged mel spectrograms of the 10 acoustic scenes, as shown in Fig. 1. The acoustic scenes of the airport, park, shopping mall, street pedestrian, and tram are narrower in terms of bandwidth but frequency components at 16kHz and above are observed in the remaining acoustic scenes.

2.2. Data Augmentation

Three augmentations using SpecAugment [4], Freq-MixStyle [5], and device impulse response (DIR) [6] are applied to the training data of the proposed models, improving system robustness and performance.

SpecAugment typically consists of three types of augmentations: time-warping, frequency masking, and time-masking. For

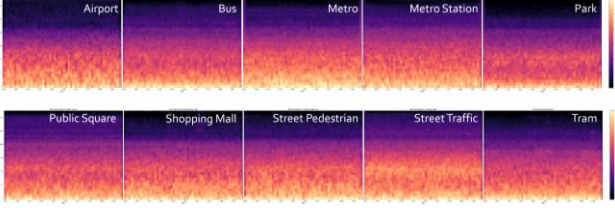


Fig. 1. Averaged mel-spectrograms of 10 acoustic scenes. Acoustic scenes of bus, metro, metro station, public square, and street traffic are found to span across a wider bandwidth.

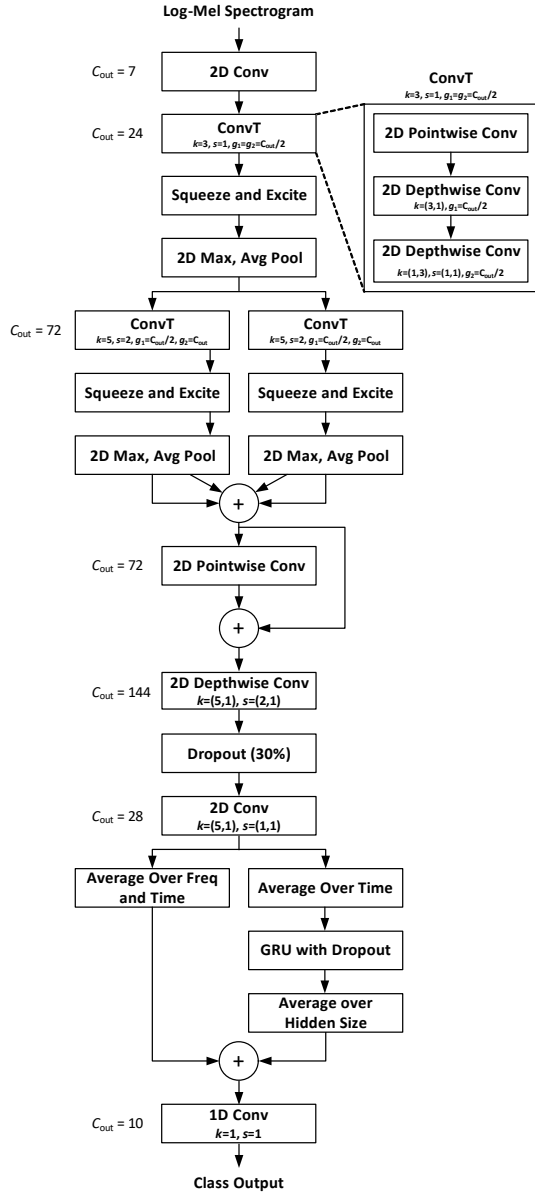


Fig. 2. Proposed GRU-CNN model. Output channels of convolution blocks are shown on the left. Input channel of model is one.

the proposed model, frequency masking has proven particularly effective. Freq-MixStyle extends MixStyle [7] to the frequency domain of the audio samples. By exposing the model to a variety

of mixed spectral properties, the model can generalize to different acoustic environments and their variations. It has been shown that models trained with Freq-MixStyle exhibit better generalization to unseen conditions, enhancing domain invariance. Models trained with DIR augmentation can also better handle variations in recording devices, leading to improved accuracy with the development dataset.

2.3. Proposed GRU-CNN Model

The architecture of the submitted model is depicted in Fig. 2. It combines 1D and 2D convolutions, DW separable convolutions, SE blocks, GRU, and hybrid pooling (average and max pooling) strategies to extract rich spectral features from log-mel spectrograms.

To reduce computational complexity, standard 2D convolutions are replaced with DW separable and pointwise convolutions. Inspired by MobileNetV2, the architecture performs pointwise convolution before DW convolution, with channel expansion applied in both stages. The DW convolution is further decomposed into two spatially separated 1D convolutions—along the time and frequency axes—to more effectively capture distinct temporal and spectral patterns. Feature map downsampling is applied in the second 1D convolution to minimize information loss during resolution reduction. The pointwise and two 1D DW convolutions are encapsulated in the ConvT block as shown in Fig. 2. SE blocks are integrated alongside both max-pooling and average-pooling operations to enhance feature representation in the early and intermediate layers.

Unlike conventional sequence models that operate along the temporal dimension, the GRU in this architecture is configured to learn patterns across the frequency axis. Specifically, the input is structured with a feature size of F (frequency bins) and a sequence length of C (channels). This design is also a consequence of the large FFT size used to compute the log-mel spectrograms. As a result, the GRU produces an output of shape (B, C, H) , where B is the batch size and H is the hidden size. These features are subsequently fused with those extracted by a parallel 1D convolutional layer, enabling complementary pattern learning across the log-mel spectrogram.

3. RESULTS AND SUBMISSION

The proposed models were trained for 150 epochs with batch size of 256 using the ADAM optimizer with the learning rate adjusted by the cosine schedule with ramp-up. The window length, hop length, FFT size, and number of mel bins are 8192, 1364, 8192, and 256, respectively. The results of the provided baseline and submitted models are summarized in Tables I and II, respectively. Compared to the baseline model, the proposed model demonstrates better classification across most classes. The first proposed model (PM1) has the architecture shown in Fig. 2, and the second proposed model (PM2) has additional channel shuffle layers between the pointwise and DW convolutions of the ConvT. The two proposed models significantly outperform the baseline model for all scenes and devices, even though no external dataset is used for training directly.

Table I Class-Wise Accuracies of Baseline in Percentage (Highest in bold)

Model	Airport	Bus	Metro	Metro Station	Park	Public Square	Shopping Mall	Street Pedestrian	Street Traffic	Tram	Macro Acc
BL	44.43	64.81	43.87	48.22	72.75	32.04	53.14	34.43	74.10	51.08	51.89
PM1	52.56	78.88	54.68	47.44	83.87	45.48	62.96	36.09	73.97	67.63	60.36
PM2	53.07	74.78	59.52	50.74	80.26	46.29	62.39	39.66	75.01	60.81	60.25

Table II Device-Wise Accuracies of Baseline in Percentage (Highest in bold)

Split	A	B	C	S1	S2	S3	S4	S5	S6	Acc
BL	63.98	55.85	59.09	48.68	48.74	52.72	48.14	47.23	42.60	51.89
PM1	69.45	61.45	63.16	58.54	57.51	62.09	58.93	59.06	53.03	60.36
PM2	69.18	60.60	65.10	57.03	56.39	62.84	58.27	59.48	53.42	60.25

4. CONCLUSIONS

In this technical report, we described the SNTL-NTU submissions to task 1 of the DCASE 2025 challenge. The proposed models are based on GRN-CNN and are trained solely on the TAU Urban Acoustic Scene 2022 Mobile development dataset, with the exception that MicRIP is used for DIR augmentation. The macro average accuracies obtained from the two proposed models are over 60% and are comparable to the submissions to the 2024 DCASE challenge.

5. ACKNOWLEDGEMENT

This research is supported by the Ministry of Education, Singapore, under its Academic Research Fund Tier 2 (MOE-T2EP20221-0014).

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